

Last problem:

Consider the following function:

$$F(x) = x I\{0 < x < 1\} + I\{x \geq 1\}, x \in \mathbf{R}$$

- (i) Is this $F(\cdot)$ non-decreasing, right-continuous ? Check.
- (ii) Define suitably a Lebesgue-Stieltjes measure μ_F on a suitable semialgebra $\mathcal{C}$. Do not have to prove $\mathcal{C}$ is a semialgebra or μ_F is a measure.
- (iii) Prove that $\sigma(\mathcal{C}) = \mathcal{B}(\mathbf{R})$. (Can use the fact that $\mathcal{B}(\mathbf{R}) = \sigma(\{\text{open sets}\}) = \sigma(O_1) = \sigma(O_2) = \sigma(O_3) = \sigma(O_4)$ as in 1.10)
- (iv) Argue that there exist a μ on $(\mathbf{R}, \mathcal{B}(\mathbf{R}))$ that extends μ_F defined above. Do not need to prove this step - just state a result for your answer.
- (v) Is this extension unique ? (i.e if μ' is another extension of μ_F , then is it true that $\mu(A) = \mu'(A)$, for **all** $A \in \mathcal{B}(\mathbf{R})$?). Justify your answer (no proof needed). What is the “common name” for this measure μ ?